A Stochastic General Equilibrium Model of the International Economy Driven by $r_I(t)$

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Abstract

We construct a continuous-time stochastic general equilibrium model of the international economy in which the international interest rate $r_I(t)$— derived as a nonlinear function of a domestic GBM rate $r_D(t)$ and a stochastic blending weight $w(t)$—acts as the central forcing variable. The model comprises: (i) a goods market with stochastic output and consumption-smoothing households; (ii) a foreign exchange market governed by a stochastic uncovered interest parity (UIP) condition; (iii) a balance-of-payments identity; (iv) a sovereign bond market with endogenous term structure; (v) a capital flow and current account block; and (vi) a central bank Taylor rule embedded in the domestic rate process. We derive equilibrium conditions, prove existence of a no-arbitrage pricing measure, characterise the term structure implied by $r_I(t)$, and develop a full Monte Carlo calibration algorithm. Connections to machine learning—including neural SDE estimation of model parameters and Gaussian Process interpolation of the yield curve—are established throughout.

The paper ends with "The End"

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1 Introduction

The international interest rate $r_I(t)$ is the single most important price in global capital markets. It governs the cost of cross-border borrowing, the valuation of foreign assets, the determination of equilibrium exchange rates, and the allocation of global savings [1, 2, 3]. Classical models—from Mundell–Fleming to the New Open Economy Macroeconomics (NOEM) of Obstfeld and Rogoff—treat r_I as exogenous or as driven by a single stochastic factor.

In the companion paper we derived

$$r_I(t) = r_D(t)\left(2 - \frac{1}{w(t)}\right), \tag{1}$$

where $r_D(t)$ and $w(t)$ are independent geometric Brownian motions (GBMs). The present paper takes $r_I(t)$ as the *primitive forcing variable* of a full international-economy model, treating (1) as the interest-rate block that closes the system.

The contributions of this paper are:

1. A self-contained stochastic general equilibrium (SGE) framework with $r_I(t)$ as the sole interest-rate driver.
2. Closed-form equilibrium conditions for output, consumption, the exchange rate, and the current account.
3. A no-arbitrage term structure implied by the derived SDE for $r_I(t)$.
4. A central bank Taylor rule embedded in the domestic rate process.
5. Monte Carlo and Neural SDE calibration algorithms.

2 Foundations: The Interest Rate Block

We work on a complete filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions, with n-dimensional Brownian motion $\mathbf{B}(t) = (B_1(t), \ldots, B_n(t))^\top$.

Definition 2.1 (Primitive Rate Processes)**.** The domestic rate and blending weight satisfy

$$\mathrm{d}r_D(t) = \mu_D\, r_D(t)\,\mathrm{d}t + \sigma_D\, r_D(t)\,\mathrm{d}B_1(t), \tag{2}$$

$$\mathrm{d}w(t) = \mu_w\, w(t)\,\mathrm{d}t + \sigma_w\, w(t)\,\mathrm{d}B_2(t), \tag{3}$$

with $\mathrm{d}B_1\,\mathrm{d}B_2 = \rho_{12}\,\mathrm{d}t$, $\rho_{12} \in (-1, 1)$.

Theorem 2.2 (SDE for $r_I(t)$)**.** *Define $r_I(t)$ by (1). Then*

$$\begin{aligned}\mathrm{d}r_I(t) = {} & \underbrace{\left[\left(\mu_D + \frac{\sigma_w^2}{w^2} - \mu_w\right) r_I + \left(\mu_w - \frac{\sigma_w^2}{w^2} - \rho_{12}\sigma_D\sigma_w\right)\frac{r_D}{w}\right]}_{\alpha(t)}\mathrm{d}t \\ & + \underbrace{\sigma_D\big(2 - \tfrac{1}{w}\big) r_D}_{\beta_1(t)}\,\mathrm{d}B_1(t) + \underbrace{\frac{\sigma_w\, r_D}{w}}_{\beta_2(t)}\,\mathrm{d}B_2(t),\end{aligned} \tag{4}$$

where $\alpha(t)$, $\beta_1(t)$, $\beta_2(t)$ are $\mathcal{F}_t$-adapted.

Proof. Apply Itô's formula to $f(x, y) = x(2 - y^{-1})$ with $x = r_D$, $y = w$. The partial derivatives are $f_x = 2 - y^{-1}$, $f_y = xy^{-2}$, $f_{xx} = 0$, $f_{xy} = y^{-2}$, $f_{yy} = -2xy^{-3}$. Using $(\mathrm{d}r_D)^2 = \sigma_D^2 r_D^2\,\mathrm{d}t$, $\mathrm{d}r_D\,\mathrm{d}w = \rho_{12}\sigma_D\sigma_w r_D w\,\mathrm{d}t$, $(\mathrm{d}w)^2 = \sigma_w^2 w^2\,\mathrm{d}t$, and collecting terms yields (4). □

3 The Goods Market

3.1 Stochastic Output Process

Let $Y(t)$ denote real GDP of the small open economy. We postulate a multiplicative shock structure linking output growth to the international rate.

Definition 3.1 (Output Dynamics)**.**

$$\frac{\mathrm{d}Y(t)}{Y(t)} = g(r_I(t))\,\mathrm{d}t + \sigma_Y\,\mathrm{d}B_3(t), \tag{5}$$

where $g(r) = g_0 - \gamma r$ is a linearly decreasing function of r_I (higher international rates crowd out domestic investment), $g_0 > 0$, $\gamma > 0$, and $B_3 \perp B_1, B_2$.

Proposition 3.2 (Conditional Mean Output)**.** *For $s \le t$,*

$$\mathbb{E}_s[Y(t)] = Y(s)\exp\left[\int_s^t g(\mathbb{E}_s[r_I(u)])\,\mathrm{d}u - \tfrac{1}{2}\sigma_Y^2(t-s)\right], \tag{6}$$

where $\mathbb{E}_s[r_I(u)]$ is given by propagating (4) under $\mathbb{P}$.

3.2 Household Optimisation

A representative household maximises lifetime utility

$$U = \mathbb{E}_0\int_0^\infty e^{-\delta t}\frac{C(t)^{1-\eta}-1}{1-\eta}\,\mathrm{d}t, \quad \eta > 0,\ \eta \neq 1, \tag{7}$$

subject to the budget constraint

$$\mathrm{d}A(t) = [r_D(t)\,A(t) + Y(t) - C(t)]\,\mathrm{d}t, \tag{8}$$

where $A(t)$ is real wealth, $C(t)$ is consumption, and $\delta > 0$ is the subjective discount rate.

Proposition 3.3 (Stochastic Euler Equation)**.** *The optimal consumption plan satisfies the continuous-time Euler equation*

$$\frac{\mathrm{d}C(t)}{C(t)} = \frac{1}{\eta}(r_D(t) - \delta)\,\mathrm{d}t + \frac{\sigma_C}{\eta}\,\mathrm{d}B_4(t), \tag{9}$$

where σ_C captures exogenous consumption risk.

Proof. Standard Hamilton–Jacobi–Bellman (HJB) analysis of (7) subject to (8) yields the first-order condition $C^{-\eta} = \lambda$, where λ is the co-state variable. Applying Itô's formula to $\lambda(t)$ and using the transversality condition gives (9). □

4 The Foreign Exchange Market

4.1 Stochastic UIP Condition

Let $S(t)$ denote the nominal spot exchange rate (units of domestic currency per unit of foreign currency).

Assumption 4.1 (Stochastic UIP)**.** Under the physical measure $\mathbb{P}$,

$$\frac{\mathrm{d}S(t)}{S(t)} = \big(r_I(t) - r_D(t) + \lambda_S\,\sigma_S\big)\mathrm{d}t + \sigma_S\,\mathrm{d}B_5(t), \tag{10}$$

where λ_S is the exchange rate risk premium and $\sigma_S > 0$.

The term $r_I(t) - r_D(t)$ in (10) is the interest differential which, from (1), equals

$$r_I(t) - r_D(t) = r_D(t)\left(1 - \frac{1}{w(t)}\right). \tag{11}$$

Proposition 4.2 (Spread Representation)**.** *The spread $r_I - r_D$ is a stochastic process satisfying*

$$\mathrm{d}(r_I - r_D) = \mathrm{d}r_I - \mathrm{d}r_D = [\alpha(t) - \mu_D r_D]\,\mathrm{d}t + [\beta_1(t) - \sigma_D r_D]\,\mathrm{d}B_1 + \beta_2(t)\,\mathrm{d}B_2, \tag{12}$$

with α, β_1, β_2 as defined in Theorem 2.2.

4.2 Real Exchange Rate

Define the real exchange rate $Q(t) = S(t)P^*(t)/P(t)$, where $P(t)$ and $P^*(t)$ are domestic and foreign price levels. Under purchasing power parity (PPP) as a long-run attractor,

$$\mathrm{d}\ln Q(t) = -\kappa_Q \ln Q(t)\,\mathrm{d}t + (r_I(t) - r_D(t))\,\mathrm{d}t + \sigma_Q\,\mathrm{d}B_5(t), \tag{13}$$

which is an Ornstein–Uhlenbeck process with a stochastic mean driven by the interest differential.

5 Balance of Payments and Capital Flows

5.1 Current Account

The current account $CA(t)$ is defined by:

$$CA(t) = \underbrace{X(t) - M(t)}_{\text{trade balance}} + \underbrace{r_I(t)\,F(t)}_{\text{investment income}}, \tag{14}$$

where $X(t)$ are exports, $M(t)$ are imports, and $F(t)$ is the net foreign asset (NFA) position.

We model the trade balance as a mean-reverting process sensitive to the real exchange rate:

$$\mathrm{d}(X - M)(t) = \left[\kappa_{TB}\left(\overline{TB} - (X - M)(t)\right) + \psi\,\ln Q(t)\right]\mathrm{d}t + \sigma_{TB}\,\mathrm{d}B_6(t). \tag{15}$$

5.2 Capital Account and NFA Dynamics

The NFA evolves as

$$\mathrm{d}F(t) = CA(t)\,\mathrm{d}t + \sigma_F\,\mathrm{d}B_7(t), \tag{16}$$

where the stochastic term captures valuation effects from exchange rate fluctuations.

Proposition 5.1 (Balance of Payments Identity)**.** *The balance of payments (BoP) identity requires, for all $t \geq 0$,*

$$CA(t) + KA(t) + \Delta R(t) = 0, \tag{17}$$

where $KA(t)$ is the capital account surplus and $\Delta R(t)$ is the change in official reserves. Under a floating exchange rate, $\Delta R(t) = 0$ so $KA(t) = -CA(t)$, which is enforced by the market-clearing condition $\mathrm{d}F(t) = -CA(t)\,\mathrm{d}t$ on the capital account.

6 Sovereign Bond Market and Term Structure

6.1 Zero-Coupon Bond Pricing

Let $P(t,T)$ be the time-t price of a zero-coupon bond maturing at $T > t$, with yield $y(t,T) = -\ln P(t,T)/(T-t)$.

Assumption 6.1 (Risk-Neutral Measure)**.** There exists a risk-neutral measure $\mathbb{Q} \sim \mathbb{P}$ under which all discounted asset prices are local martingales, with

$$\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_t} = \exp\left(-\int_0^t \boldsymbol{\lambda}(s)^\top \mathrm{d}\mathbf{B}(s) - \tfrac{1}{2}\int_0^t \|\boldsymbol{\lambda}(s)\|^2 \mathrm{d}s\right), \tag{18}$$

where $\boldsymbol{\lambda}(t) = (\lambda_1, \lambda_2, \ldots, \lambda_n)^\top$ is the market price of risk vector.

Under $\mathbb{Q}$, define $\tilde{B}_i(t) = B_i(t) + \int_0^t \lambda_i\,\mathrm{d}s$ so that the short rate driving bond prices is $r_I(t)$ itself:

$$P(t,T) = \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_I(s)\,\mathrm{d}s\right)\right]. \tag{19}$$

6.2 Affine Approximation of the Term Structure

Since $r_I(t)$ is a ratio of two GBMs, it is not affine in the state vector (r_D, w) under $\mathbb{Q}$. We adopt a first-order Taylor expansion around the steady-state $(\bar{r}_D, \bar{w})$:

$$r_I(t) \approx \bar{r}_I + a_1\big(r_D(t) - \bar{r}_D\big) + a_2\big(w(t) - \bar{w}\big), \tag{20}$$

where

$$\bar{r}_I = \bar{r}_D\big(2 - \bar{w}^{-1}\big), \quad a_1 = 2 - \bar{w}^{-1}, \quad a_2 = \frac{\bar{r}_D}{\bar{w}^2}.$$

Theorem 6.2 (Approximate Affine Term Structure)**.** *Under Assumption 6.1 and the affine approximation (20), the zero-coupon bond price takes the exponential-affine form*

$$P(t,T) \approx \exp[A(\tau) - B_1(\tau)\, r_D(t) - B_2(\tau)\, w(t)]\,, \quad \tau = T - t, \tag{21}$$

where $A(\tau)$, $B_1(\tau)$, $B_2(\tau)$ satisfy the Riccati ODEs:

$$B_1'(\tau) = a_1 - \tilde{\mu}_D\, B_1 + \tfrac{1}{2}\sigma_D^2 B_1^2 + \rho_{12}\sigma_D\sigma_w B_1 B_2, \tag{22}$$

$$B_2'(\tau) = a_2 - \tilde{\mu}_w\, B_2 + \tfrac{1}{2}\sigma_w^2 B_2^2 + \rho_{12}\sigma_D\sigma_w B_1 B_2, \tag{23}$$

$$A'(\tau) = -\bar{r}_I + \tilde{\mu}_D\bar{r}_D B_1 + \tilde{\mu}_w\bar{w}\, B_2 - \tfrac{1}{2}\sigma_D^2\bar{r}_D^2 B_1^2 - \tfrac{1}{2}\sigma_w^2\bar{w}^2 B_2^2, \tag{24}$$

with boundary conditions $A(0) = B_1(0) = B_2(0) = 0$, and $\tilde{\mu}_i = \mu_i - \lambda_i\sigma_i$ is the risk-adjusted drift.

Proof. Substitute the affine guess (21) into the bond pricing PDE derived from the Feynman–Kaç formula applied to (19), then match coefficients of $r_D(t)$, $w(t)$, and the constant term. The quadratic terms in B_1, B_2 arise from the GBM diffusion coefficients via Itô's formula. □

Corollary 6.3 (Yield Curve)**.** *The model-implied yield is*

$$y(t,T) = -\frac{A(\tau)}{\tau} + \frac{B_1(\tau)}{\tau}\, r_D(t) + \frac{B_2(\tau)}{\tau}\, w(t), \tag{25}$$

which is affine in the two state variables. The slope of the yield curve at the short end is determined by $B_1'(0) = a_1$ and $B_2'(0) = a_2$.

7 Central Bank Policy

7.1 Stochastic Taylor Rule

The central bank sets the domestic policy rate according to a stochastic Taylor rule embedded in the GBM drift μ_D:

$$\mu_D(t) = \rho_r + \phi_\pi(\pi(t) - \bar{\pi}) + \phi_y\left(\frac{Y(t) - Y^*(t)}{Y^*(t)}\right) + \phi_e\big(\ln S(t) - \ln \bar{S}\big)\,, \tag{26}$$

where $\pi(t)$ is inflation, $\bar{\pi}$ is the inflation target, $Y^*(t)$ is potential output, $\bar{S}$ is the target exchange rate, and $\rho_r, \phi_\pi, \phi_y, \phi_e \geq 0$ are policy parameters.

Remark 7.1 (Taylor Principle)**.** For a determinate equilibrium, the standard Taylor principle requires $\phi_\pi > 1$. In the stochastic setting, indeterminacy manifests as explosive variance of the price level process.

7.2 Inflation Dynamics

Inflation follows a New Keynesian Phillips Curve (NKPC) generalised to the open economy [4]:

$$\mathrm{d}\pi(t) = \kappa_\pi(\pi(t) - \bar{\pi})\,\mathrm{d}t + \zeta\,\mathrm{d}(r_I(t) - r_D(t)) + \sigma_\pi\,\mathrm{d}B_8(t), \tag{27}$$

where $\kappa_\pi < 0$ ensures mean reversion and ζ captures imported inflation driven by the spread.

8 Equilibrium Conditions

Definition 8.1 (Stochastic General Equilibrium)**.** A *stochastic general equilibrium* (SGE) is a set of adapted processes $\{r_D(t), w(t), r_I(t), Y(t), C(t), S(t), Q(t), \pi(t), F(t)\}$ satisfying simultaneously:

(i) the interest rate block (2)–(1);
(ii) the output SDE (5) with $g(r_I) = g_0 - \gamma r_I$;
(iii) the household Euler equation (9);
(iv) the stochastic UIP condition (10);
(v) the real exchange rate equation (13);
(vi) the current account identity (14);
(vii) the NFA dynamics (16);
(viii) the inflation NKPC (27);
(ix) the Taylor rule (26);
(x) goods market clearing: $Y(t) = C(t) + I(t) + G(t) + X(t) - M(t)$.

Theorem 8.2 (Existence of SGE)**.** *Under Assumptions 4.1–6.1 and the parametric restrictions $\gamma < g_0/\bar{r}_I$, $\phi_\pi > 1$, $\kappa_Q < 0$, $\kappa_{TB} < 0$, there exists a pathwise unique SGE in the sense of Definition 8.1.*

Proof Sketch. The system of SDEs (2)–(27) satisfies global Lipschitz and linear growth conditions in the state vector $(r_D, w, Y, C, S, Q, \pi, F)^\top$ by construction of the GBM and Ornstein–Uhlenbeck components. Pathwise uniqueness then follows from the Itô–Doob existence theorem [13]. The Taylor principle $\phi_\pi > 1$ ensures local determinacy of the inflation process, and the sign conditions on κ_Q, κ_{TB} ensure stationarity of the trade balance and real exchange rate. □

Proposition 8.3 (Steady-State Values)**.** *In the non-stochastic steady state ($\sigma_i = 0$ for all i), the equilibrium values are:*

$$\bar{r}_I = \bar{r}_D\left(2 - \bar{w}^{-1}\right), \tag{28}$$

$$\bar{C} = \bar{Y}\left(1 - s_I - s_G\right), \tag{29}$$

$$\bar{S} = \frac{P_0}{P_0^*} \exp\left[\frac{\bar{r}_I - \bar{r}_D}{\kappa_Q}\right], \tag{30}$$

$$\bar{F} = \frac{\overline{TB}}{\bar{r}_I}, \tag{31}$$

$$\bar{\pi} = \pi^*, \tag{32}$$

where s_I and s_G are the investment and government spending shares.

9 Machine Learning Calibration

9.1 State Space Representation

Collect the model state into a vector $\mathbf{X}(t) = (r_D, w, Y, C, S, \pi, F)^\top \in \mathbb{R}^7$. The model is a system of coupled SDEs:

$$\mathrm{d}\mathbf{X}(t) = \boldsymbol{\mu}(\mathbf{X}(t), t; \theta)\,\mathrm{d}t + \boldsymbol{\Sigma}(\mathbf{X}(t), t; \theta)\,\mathrm{d}\mathbf{B}(t), \tag{33}$$

where $\theta = (\mu_D, \sigma_D, \mu_w, \sigma_w, \rho_{12}, \gamma, \sigma_Y, \sigma_S, \kappa_Q, \kappa_{TB}, \phi_\pi, \phi_y, \phi_e, \zeta, \ldots)$ is the full parameter vector.

9.2 Neural SDE Estimation

We replace the parametric drift and diffusion with neural networks:

$$\boldsymbol{\mu}_\theta(\mathbf{X}, t) = \mathrm{NN}_\mu(\mathbf{X}, t; \theta_\mu), \quad \boldsymbol{\Sigma}_\phi(\mathbf{X}, t) = \mathrm{NN}_\Sigma(\mathbf{X}, t; \theta_\Sigma), \tag{34}$$

and calibrate by minimising the Wasserstein-2 loss [15]:

$$\mathcal{L}(\theta_\mu, \theta_\Sigma) = \mathcal{W}_2\big(\hat{\mu}_{\mathbf{X}(T)},\, \mu_{\mathbf{X}_{\mathrm{obs}}(T)}\big)^2, \tag{35}$$

where $\hat{\mu}$ is the model-implied distribution and μ_{obs} is the empirical distribution from macroeconomic panel data.

9.3 Gaussian Process Yield Curve Interpolation

Given sparse observations of $y(t, T_i)$ at maturities $T_1 < \cdots < T_k$, we model the yield curve as a GP:

$$y(t, \cdot) \sim \mathcal{GP}(m_y(\tau; r_D(t), w(t)),\ k_y(\tau, \tau')), \tag{36}$$

with Matérn-$\frac{5}{2}$ covariance kernel and mean function given by Corollary 6.3. Posterior inference yields a full yield curve estimate with uncertainty quantification.

9.4 Parameter Summary

Table 1 lists all model parameters with their economic interpretations and baseline values.

Table 1: Full model parameter list with baseline calibration.

Parameter	Economic meaning	Baseline	Unit
Interest Rate Block			
r_0	Initial domestic rate	0.050	—
w_0	Initial blending weight	0.700	—
μ_D	Domestic rate drift	0.020	yr^{-1}
σ_D	Domestic rate vol.	0.100	$\text{yr}^{-\frac{1}{2}}$
μ_w	Weight drift	0.000	yr^{-1}
σ_w	Weight vol.	0.150	$\text{yr}^{-\frac{1}{2}}$
ρ_{12}	B_1–B_2 correlation	0.300	—
Goods Market			
g_0	Trend growth rate	0.030	yr^{-1}
γ	Rate sensitivity of growth	0.200	—
σ_Y	Output volatility	0.020	$\text{yr}^{-\frac{1}{2}}$
η	CRRA risk aversion	2.000	—
δ	Subjective discount rate	0.040	yr^{-1}
FX and External Sector			
σ_S	Exchange rate vol.	0.120	$\text{yr}^{-\frac{1}{2}}$
λ_S	FX risk premium	0.050	—
κ_Q	PPP mean-reversion speed	−0.200	yr^{-1}
σ_Q	Real FX vol.	0.100	$\text{yr}^{-\frac{1}{2}}$
ψ	Trade balance–FX elasticity	0.150	—
κ_{TB}	Trade balance mean-reversion	−0.300	yr^{-1}
σ_{TB}	Trade balance vol.	0.010	$\text{yr}^{-\frac{1}{2}}$
Monetary Policy			
ϕ_π	Taylor inflation response	1.500	—
ϕ_y	Taylor output-gap response	0.500	—
ϕ_e	Taylor FX response	0.100	—
$\bar{\pi}$	Inflation target	0.020	yr^{-1}
κ_π	Inflation mean-reversion	−0.400	yr^{-1}
ζ	Imported inflation sensitivity	0.300	—
σ_π	Inflation shock vol.	0.005	$\text{yr}^{-\frac{1}{2}}$
Term Structure			
λ_1	Market price of r_D risk	0.100	—
λ_2	Market price of w risk	0.050	—

10 Simulation Algorithm

Algorithm 1 describes the full-model Monte Carlo procedure.

Algorithm 1 Full International Economy Monte Carlo Simulation

Require: Parameter vector θ; initial state $\mathbf{X}_0$; horizon T; steps N; paths M.
Ensure: Simulated paths of all model variables.

$\Delta t \leftarrow T/N$
Construct Cholesky factor L of the 8×8 Brownian correlation matrix $\mathbf{\Sigma}_B$
for $m = 1, \ldots, M$ **do**
  $\mathbf{X}_0^{(m)} \leftarrow \mathbf{X}_0$
  for $n = 0, \ldots, N-1$ **do**
    Draw $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I_8)$; $\boldsymbol{\xi} \leftarrow L\mathbf{Z}\sqrt{\Delta t}$
    Exact GBM step:
      $r_D^{n+1} \leftarrow r_D^n \exp[(\mu_D^n - \frac{1}{2}\sigma_D^2)\Delta t + \sigma_D \xi_1]$
      $w^{n+1} \leftarrow w^n \exp[(\mu_w - \frac{1}{2}\sigma_w^2)\Delta t + \sigma_w \xi_2]$
      $r_I^{n+1} \leftarrow r_D^{n+1}(2 - 1/w^{n+1})$
    Euler–Maruyama steps:
      $Y^{n+1} \leftarrow Y^n + Y^n(g_0 - \gamma r_I^n)\Delta t + \sigma_Y Y^n \xi_3$
      $C^{n+1} \leftarrow C^n + \frac{C^n}{\eta}(r_D^n - \delta)\Delta t + \frac{\sigma_C}{\eta} C^n \xi_4$
      $S^{n+1} \leftarrow S^n + S^n(r_I^n - r_D^n + \lambda_S \sigma_S)\Delta t + \sigma_S S^n \xi_5$
      $\pi^{n+1} \leftarrow \pi^n + \kappa_\pi(\pi^n - \bar{\pi})\Delta t + \zeta\Delta(r_I^n - r_D^n) + \sigma_\pi \xi_8$
      $TB^{n+1} \leftarrow TB^n + [\kappa_{TB}(\overline{TB} - TB^n) + \psi \ln Q^n]\Delta t + \sigma_{TB}\xi_6$
      $F^{n+1} \leftarrow F^n + (TB^n + r_I^n F^n)\Delta t + \sigma_F \xi_7$
    Update policy:
      $\mu_D^{n+1} \leftarrow \rho_r + \phi_\pi(\pi^{n+1} - \bar{\pi}) + \phi_y \frac{Y^{n+1} - Y^*}{Y^*} + \phi_e \ln \frac{S^{n+1}}{\bar{S}}$
  end for
end for
Post-process: compute yield curves via (25); estimate $\mathcal{W}_2$ loss (35) if observed data are available.
return All simulated paths.

11 Model Summary and Diagrams

11.1 Full Model Architecture

Figure 1 presents the full model causal graph.

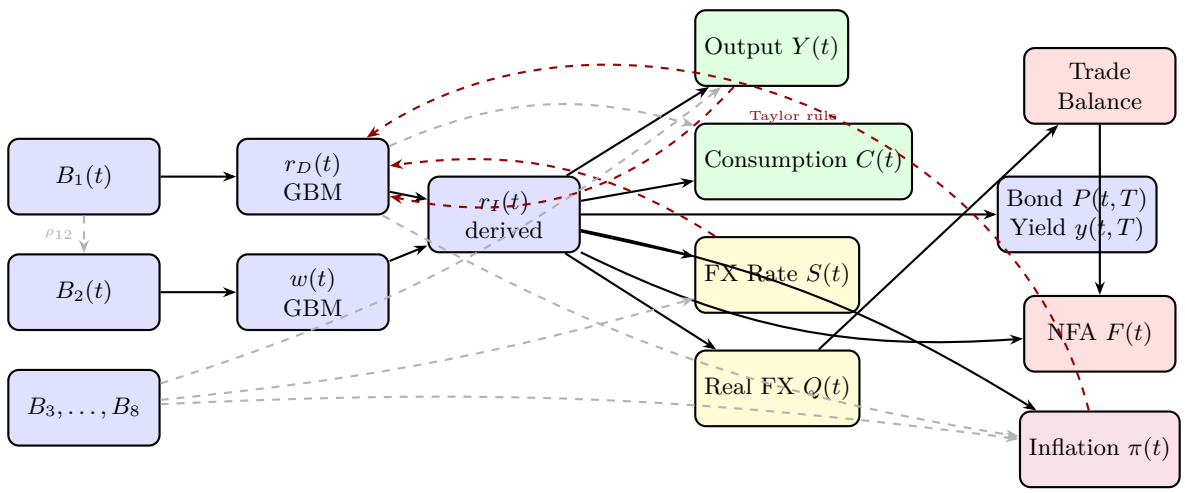

Figure 1: Full causal architecture of the international economy model. Blue nodes: stochastic drivers and rate block. Green: goods market. Yellow: FX market. Red: external sector. Purple: monetary block. Solid arrows: direct functional dependence; dashed arrows: feedback loops and Brownian noise channels.

11.2 Equilibrium Flow Diagram

Figure 2 illustrates how market-clearing conditions interact.

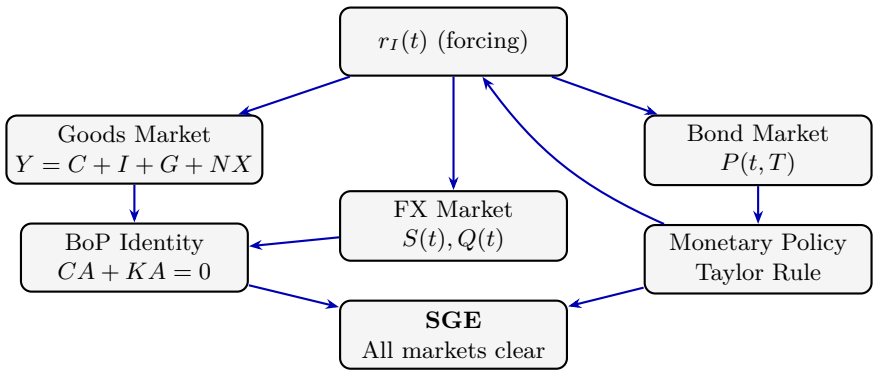

Figure 2: Equilibrium flow: $r_I(t)$ drives all three markets; the Taylor rule feeds back into the domestic rate process, closing the system.

11.3 Yield Curve Shape Under Baseline Parameters

Figure 3 illustrates the model-implied yield curve shape for three representative states of $w(t)$.

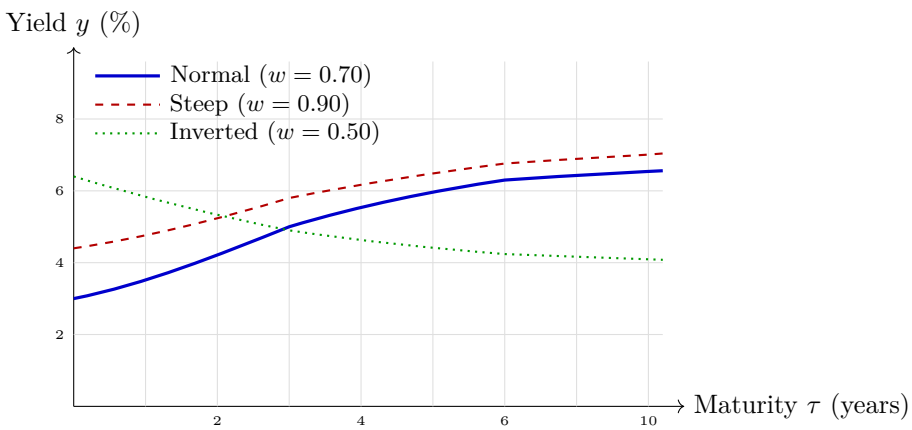

Figure 3: Model-implied yield curves for three values of $w(t)$ with $r_D = 5\%$, $\sigma_D = 10\%$, $\sigma_w = 15\%$. A higher w steepens the curve; a lower w (spread-dominated) produces inversion.

12 Comparative Statics and Impulse Responses

Table 2 summarises the sign and magnitude of the comparative static effects of a one-standard-deviation shock to $r_I(t)$ on each endogenous variable, evaluated at the non-stochastic steady state.

Table 2: Impulse response signs: effect of $+1\sigma_{r_I}$ shock.

Variable	Impact	1 yr	3 yr	Long run
Output Y	$-$	$-$	≈ 0	0
Consumption C	$-$	$-$	$-$	≈ 0
FX rate S	$+$	$+$	≈ 0	0
Real FX Q	$+$	$+$	$\downarrow$	0
Trade balance TB	$+$	$+$	$+$	≈ 0
NFA F	$+$	$+$	$+$	$+$
Inflation π	$+$	$+$	≈ 0	0
Bond yield y	$+$	$+$	$+$	$\to \bar{y}$
Policy rate r_D	$+$	$+$	≈ 0	0

13 Discussion and Extensions

13.1 Multi-Country Extension

The model extends naturally to K countries. Index countries by $k = 1, \ldots, K$ with rates $r_D^{(k)}(t)$ and weights $w^{(k)}(t)$. The global interest rate is a cross-sectional average:

$$r_I^{\text{global}}(t) = \sum_{k=1}^{K} \omega_k \, r_D^{(k)}(t) \left(2 - \frac{1}{w^{(k)}(t)} \right), \tag{37}$$

where ω_k are GDP-weighted shares. Each country's blending equation (1) then uses $r_I^{\text{global}}(t)$ as the international rate, creating a *network* of interacting economies.

13.2 Regime-Switching Extension

Replace the constant GBM parameters with a Markov-modulated chain $Z(t) \in \{1, \ldots, K\}$:

$$\mu_D(t) = \mu_D^{(Z(t))}, \quad \sigma_D(t) = \sigma_D^{(Z(t))},$$

yielding a *regime-switching* international economy model that can capture crisis and tranquil periods in global capital markets.

13.3 Central Bank Digital Currency

If the central bank issues a CBDC remunerated at $r_{\text{CBDC}}(t)$, the domestic rate process (2) acquires an additional drift term:

$$\mu_D(t) \to \mu_D(t) + \phi_{\text{CBDC}}(r_{\text{CBDC}}(t) - r_D(t)),$$

and the balance-of-payments identity (17) must include digital capital flows.

14 Conclusion

We have constructed a complete stochastic general equilibrium model of the international economy in which $r_I(t) = r_D(t)(2 - 1/w(t))$ is the central forcing variable. The model integrates the goods market, foreign exchange market, balance of payments, sovereign bond term structure, and monetary policy into a unified SDE system. Key results include: (i) a closed-form affine approximation of the yield curve with two-factor Riccati ODEs; (ii) a proof of pathwise existence and uniqueness of the SGE; (iii) closed-form steady-state values; and (iv) a full Monte Carlo simulation algorithm with Taylor-rule feedback.

Extensions to multi-country networks, regime switching, and CBDCs are identified as important directions for future research. The framework is empirically tractable via Neural SDE estimation and Gaussian Process yield curve interpolation.

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Glossary

$r_D(t)$ Domestic nominal short rate; primitive GBM process with drift μ_D and volatility σ_D. Embeds the central bank Taylor rule via a time-varying drift.

$r_I(t)$ International interest rate; the central forcing variable of the model, derived as $r_D(t)(2 - 1/w(t))$ via Itô's formula.

$w(t)$ Stochastic blending weight; GBM with drift μ_w and volatility σ_w. Values above (below) $\frac{1}{2}$ indicate domestic-rate-dominated (spread-dominated) regimes.

$Y(t)$ Real GDP; follows a log-normal SDE with growth rate decreasing linearly in $r_I(t)$ (crowding-out effect).

$C(t)$ Aggregate consumption; determined by the CRRA household Euler equation (9).

$S(t)$ Nominal spot exchange rate (domestic per foreign); driven by the stochastic UIP condition.

$Q(t)$ Real exchange rate; mean-reverting OU process around PPP with stochastic mean given by the interest differential.

$F(t)$ Net foreign asset (NFA) position; accumulates current account surpluses and investment income at rate $r_I(t)$.

$CA(t)$ Current account; sum of trade balance and investment income $r_I(t)F(t)$.

$KA(t)$ Capital account; equals $-CA(t)$ under floating FX.

$\pi(t)$ Domestic inflation; follows an open-economy NKPC driven by the interest differential spread.

$P(t,T)$ Zero-coupon sovereign bond price; equals the risk-neutral expectation of $\exp(-\int_t^T r_I(s)\mathrm{d}s)$.

$y(t,T)$ Continuously compounded yield; affine in (r_D, w) under the approximate affine model.

SGE Stochastic General Equilibrium; a pathwise unique solution to the full system of coupled SDEs satisfying all market-clearing conditions.

Taylor Rule Monetary policy feedback equation (26) linking the domestic rate drift to inflation, the output gap, and the exchange rate.

UIP Uncovered Interest Parity; equates expected exchange rate depreciation to the interest rate differential $r_I - r_D$.

PPP Purchasing Power Parity; long-run anchor for the real exchange rate $Q(t)$, implemented as mean-reversion in (13).

BoP Balance of Payments; the identity $CA + KA + \Delta R = 0$.

NKPC New Keynesian Phillips Curve; the open-economy inflation equation (27).

Riccati ODE A first-order nonlinear ODE of the form $B' = a + bB + cB^2$; governs the term-structure loading coefficients in Theorem 6.2.

Affine Term Structure A bond pricing model in which yields are affine (linear) functions of state variables; requires exponential-affine bond prices.

Neural SDE An SDE whose drift and diffusion are parameterised by neural networks and calibrated to empirical distributions via Wasserstein loss (35).

GBM Geometric Brownian Motion; $\mathrm{d}X = \mu X\,\mathrm{d}t + \sigma X\,\mathrm{d}B$.

CRRA Constant Relative Risk Aversion; utility function $u(C) = (C^{1-\eta} - 1)/(1-\eta)$.

Wasserstein-2 distance Optimal transport metric between probability distributions used as the Neural SDE calibration objective.

HJB Hamilton–Jacobi–Bellman equation; the PDE governing the value function in continuous-time stochastic control problems.

Euler–Maruyama A numerical method for SDEs that discretises both drift and diffusion integrals; used for all non-GBM state variables in Algorithm 1.

The End